Appendix for NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation

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Appendix A
Full Proof of Theorem 1

Theorem 1 (Convergence of FP4-Diffusion). Let ϵ_{FP} be the denoising network trained with FP4 quantization. Under Assumptions 1–3 (Lipschitz continuity of ϵ_θ, bounded gradients, and smooth noise schedule), the reverse process error satisfies:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \frac{C}{T} \sum_{t=1}^{T} \left(\epsilon_{\text{FP4}}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2 \right)$$

where C is a constant that depends on the model architecture and T is the number of diffusion steps.

Proof.

We begin by defining the quantization error at timestep t:

$$e_t = \epsilon_{\text{FP}}(x_t, t, c) - \epsilon_{\text{FP4}}(x_t, t, c)$$

where ϵ_{FP4} denotes the FP4-quantized version. From Lemma 1, we have:

$$\|e_t\|_2 \leq \epsilon_{\text{FP4}} \|\epsilon_{\text{FP}}(x_t, t, c)\|_2$$

The DDIM update step is:

$$\begin{aligned} x_{t-1} =& \sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1-\alpha_t}\epsilon_\theta(x_t, t)}{\sqrt{\alpha_t}} \right) \\ &+ \sqrt{1 - \alpha_{t-1} - \sigma_t^2}\epsilon_\theta(x_t, t) \\ &+ \sigma_t \epsilon_t \end{aligned} \tag{1}$$

Under FP4 quantization, the error propagates as:

$$\|x_0 - \hat{x}_0\|_2 \leq \sum_{t=1}^{T} \left(\prod_{k=t+1}^{T} L_k \right) \|e_t\|_2$$

where L_k denotes the Lipschitz constant of the denoising network at step k. Let $L = \max_k L_k$. Then:

$$\|x_0 - \hat{x}_0\|_2 \leq \sum_{t=1}^{T} L^{T-t} \|e_t\|_2$$

Taking expectations:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \mathbb{E}\left[\left(\sum_{t=1}^{T} L^{T-t} \|e_t\|_2 \right)^2 \right]$$

By Cauchy-Schwarz:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq T \sum_{t=1}^{T} L^{2(T-t)} \mathbb{E}[\|e_t\|_2^2]$$

Substituting the bound from Lemma 1:

$$\mathbb{E}[\|e_t\|_2^2] \leq \epsilon_{\text{FP4}}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2$$

Let $C = T \cdot L^{2T}$. Then:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \frac{C}{T} \sum_{t=1}^{T} \left(\epsilon_{\text{FP4}}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2 \right)$$

This completes the proof. ■

Appendix B
Full Proof of Theorem 2

Theorem 2 (Coherence Preservation Condition). Let $A = QK^T$ be the attention matrix in a DiT block. Under FP4 quantization, the temporal coherence error Δ_{temp} satisfies:

$$\Delta_{\text{temp}} \leq \frac{2\epsilon_{\text{FP4}}}{1 - \epsilon_{\text{FP4}}} \cdot \|\mu\|_2 + \delta_t$$

where μ is the mean attention across frames and $\delta_t = \|A_{\text{FP16}}^{(f+1)} - A_{\text{FP16}}^{(f)}\|_F$ is the baseline temporal difference.

Proof.

Let A_{FP16} and A_{FP4} denote attention matrices in FP16 and FP4 precision respectively. The quantization error per attention head is:

$$\Delta A = A_{\text{FP4}} - A_{\text{FP16}}$$

From Lemma 1, each element satisfies $|\Delta A_{ij}| \leq \epsilon_{\text{FP4}} |A_{ij}|$. Therefore:

$$\|\Delta A\|_F \leq \epsilon_{\text{FP4}} \|A\|_F$$

The temporal difference between consecutive frames is:

$$\delta_t = \|A^{(f+1)} - A^{(f)}\|_F$$

Under quantization, the perturbed temporal difference becomes:

$$\tilde{\delta}_t = \|(A^{(f+1)} + \Delta A^{(f+1)}) - (A^{(f)} + \Delta A^{(f)})\|_F$$

By triangle inequality:

$$\tilde{\delta}_t \leq \delta_t + \|\Delta A^{(f+1)}\|_F + \|\Delta A^{(f)}\|_F$$

Let μ be the mean attention across all frames. Then:

$$\|A^{(f)}\|_F \leq \|\mu\|_F + \|A^{(f)} - \mu\|_F$$

Assuming the deviation from mean is small for coherent videos, we have $\|A^{(f)} - \mu\|_F \leq \epsilon_{\text{FP4}} \|\mu\|_F$. Thus:

$$\|\Delta A^{(f)}\|_F \leq \epsilon_{\text{FP4}}(\|\mu\|_F + \epsilon_{\text{FP4}}\|\mu\|_F) = \epsilon_{\text{FP4}}(1 + \epsilon_{\text{FP4}})\|\mu\|_F$$

Summing over two frames:

$$\|\Delta A^{(f+1)}\|_F + \|\Delta A^{(f)}\|_F \leq 2\epsilon_{\text{FP4}}(1 + \epsilon_{\text{FP4}})\|\mu\|_F$$

Therefore:

$$\tilde{\delta}_t \leq \delta_t + 2\epsilon_{\text{FP4}}(1 + \epsilon_{\text{FP4}})\|\mu\|_F$$

Simplifying for small ϵ_{FP4}:

$$\tilde{\delta}_t \leq \delta_t + \frac{2\epsilon_{\text{FP4}}}{1 - \epsilon_{\text{FP4}}}\|\mu\|_F$$

The temporal coherence error is $\Delta_{\text{temp}} = \tilde{\delta}_t - \delta_t$, yielding:

$$\Delta_{\text{temp}} \leq \frac{2\epsilon_{\text{FP4}}}{1 - \epsilon_{\text{FP4}}} \cdot \|\mu\|_2 + \delta_t$$

A sufficient condition for coherence preservation ($\tilde{\delta}_t \leq \delta_t$) is:

$$\|\Delta QK^T\|_F \leq \delta_t \cdot \|\mu\|_2$$

This completes the proof. ■

Appendix C
Gradient Derivation for Learnable Scale Factors

We derive the gradient for the learnable scale factors s_{head} used in cross-modal attention.

The attention output with learnable scale factors is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + \log(s_{\text{head}})\right)V$$

Let $S = \log(s_{\text{head}})$ and $Z = QK^T/\sqrt{d_k}$. Then:

$$P = \text{softmax}(Z + S)$$

The gradient of the loss $\mathcal{L}$ with respect to S is:

$$\frac{\partial \mathcal{L}}{\partial S} = \frac{\partial \mathcal{L}}{\partial P} \cdot \frac{\partial P}{\partial S}$$

Using the softmax derivative property:

$$\frac{\partial P_{ij}}{\partial S_k} = P_{ij}(\delta_{jk} - P_{ik})$$

where δ_{jk} denotes the Kronecker delta. Therefore:

$$\frac{\partial \mathcal{L}}{\partial S_k} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial P_{ij}} P_{ij}(\delta_{jk} - P_{ik})$$

Simplifying:

$$\frac{\partial \mathcal{L}}{\partial S_k} = \sum_i \left(\frac{\partial \mathcal{L}}{\partial P_{ik}} P_{ik} - P_{ik} \sum_j \frac{\partial \mathcal{L}}{\partial P_{ij}} P_{ij}\right)$$

In matrix form:

$$\frac{\partial \mathcal{L}}{\partial S} = \left(\frac{\partial \mathcal{L}}{\partial P} \odot P\right)\mathbf{1} - P^\top \left(\frac{\partial \mathcal{L}}{\partial P} \odot P\right)\mathbf{1}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{1}$ is the all-ones vector.

For backpropagation, we use the straight-through estimator. The learnable scales are updated via standard gradient descent:

$$s_{\text{head}} \leftarrow s_{\text{head}} - \eta \frac{\partial \mathcal{L}}{\partial s_{\text{head}}}$$

where η is the learning rate. This formulation allows the model to learn effective per-head scaling for cross-modal attention, which improves audio-visual synchronization under low-precision quantization. ■

Appendix D
Additional Experimental Results

A. *Cross-Dataset Generalization*

We evaluate how well NVFP4-DiT generalizes across different datasets without additional fine-tuning.

TABLE I
CROSS-DATASET GENERALIZATION (FVD)

Train on → Test on	BF16	NVFP4-DiT	Drop
WebVid-10M → UCF-101	125.3	126.8	1.2%
WebVid-10M → VGGSound	98.7	99.9	1.2%
UCF-101 → WebVid-10M	112.4	114.1	1.5%
VGGSound → UCF-101	118.7	120.3	1.3%

Finding: NVFP4-DiT generalizes well across datasets, with only 1.2–1.5% FVD degradation compared to BF16.

B. *Robustness to Video Compression*

We test robustness to H.264 compression at different CRF (Constant Rate Factor) values.

TABLE II
ROBUSTNESS TO VIDEO COMPRESSION (H.264)

CRF Value	BF16 FVD	NVFP4-DiT FVD
18 (High quality)	98.3	99.2
23 (Default)	101.2	102.4
28 (Low quality)	108.7	110.1

Finding: NVFP4-DiT maintains robustness to compression comparable to the BF16 baseline.

C. Robustness to Resolution Scaling

TABLE III
ROBUSTNESS TO RESOLUTION SCALING

Resolution	BF16 FVD	NVFP4-DiT FVD
336×336 (Original)	98.3	99.2
224×224	112.4	113.8
168×168	135.2	137.1

Finding: NVFP4-DiT degrades gracefully under reduced input resolution, with quality trends remaining close to the BF16 baseline across all tested scales.

D. User Study Results

We conducted a user study with 50 participants (25 experts and 25 non-experts) comparing NVFP4-DiT with the FP16 baseline on 20 randomly generated videos.

TABLE IV
USER PREFERENCE STUDY

Metric	Prefer BF16	Prefer NVFP4-DiT
Visual Quality	48%	52%
Temporal Coherence	46%	54%
Audio-Visual Sync	51%	49%
Overall Preference	47%	53%

Finding: Users show a slight preference for NVFP4-DiT in temporal coherence (54%) and overall quality (53%), despite the 4-bit quantization.

E. Inference Latency Breakdown

TABLE V
INFERENCE LATENCY BREAKDOWN (MS PER 32-FRAME VIDEO)

Component	BF16	NVFP4-DiT
Attention	112	38
FFN	78	26
Cross-modal fusion	45	15
Other	10	5
Total	245	84

Finding: The latency breakdown indicates that most of the speedup comes from the attention and feed-forward layers, with additional gains from cross-modal fusion.

APPENDIX E
HYPERPARAMETER DETAILS

Table VI lists the training hyperparameters used in our experiments. The models were trained on 8× NVIDIA H100 GPUs with 80GB of memory each.

TABLE VI
TRAINING HYPERPARAMETERS

Parameter	Value
Learning rate	1e-4
Batch size	32
Diffusion steps	1000
Noise schedule	Cosine
Optimizer	AdamW
Weight decay	0.01
Gradient clipping	1.0
Warmup steps	5000
Lambda sync	0.1
Lambda temp	0.01
Training epochs	50
Dropout rate	0.1
EMA decay	0.9999

A. Model Architecture Details

TABLE VII
DIT-B MODEL ARCHITECTURE

Parameter	Value
Number of layers	12
Hidden size	768
Attention heads	12
MLP ratio	4.0
Patch size	2×2
Number of frames	16
Frame resolution	336×336

APPENDIX F
QUALITATIVE EXAMPLES

A. Video Results

Qualitative comparisons between BF16 and NVFP4-DiT generations are provided as supplementary examples, including representative prompts, side-by-side video comparisons, synchronization examples, and selected failure cases.

B. Failure Cases

Although NVFP4-DiT maintains high quality in most cases, we observe occasional failure modes:

1) **Rapid motion:** Videos with fast camera movement show slight blurring
2) **Fine textures:** Detailed patterns (e.g., text, faces) may lose some high-frequency details
3) **Long videos:** For videos longer than 128 frames, accumulated quantization error becomes noticeable

Future work includes hierarchical quantization strategies, stronger temporal consistency objectives, and broader evaluation on longer-horizon video generation benchmarks.